

CO2 – Individual Assignment

1. HEADER

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Course Code & CO Reference: CSA0920 – CO2

Assignment Set: Direct-Descriptive

Question No.: Q30

Assigned Question

Q30. Generic Box Class

Write a Java program that defines a generic class named `Box<T>`, capable of storing a single value of any data type and returning it later. Demonstrate the flexibility of this generic class by creating separate `Box` instances that store an `Integer`, a `String`, and a `Double` respectively, and display the value retrieved from each.

Concepts: Generics, Generic Class.

GitHub Repository Link:

https://github.com/SasiKumarS254/CSA0920-Programming-in-JAVA/blob/main/30_CO2_Q30

Submission Date: 1/8/2026

2. PROBLEM UNDERSTANDING

The objective of this program is to understand how Java Generics work by creating a generic class that can store different types of data. Instead of writing separate classes for each data type, one generic class is used to store and retrieve `Integer`, `String` and `Double` values.

3. APPROACH / DESIGN NOTES

- Created a generic class named Box<T>.
- Used a generic variable to store a value.
- Used Scanner to read values from the user.
- Created separate objects for Integer, String and Double.
- Used getValue() method to retrieve and display the stored values.

4. SOURCE CODE

```
import java.util.Scanner;

class Box<T> {

    private T value;

    public void setValue(T value) {
        this.value = value;
    }

    public T getValue() {
        return value;
    }
}

public class GenericBox {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Box<Integer> integerBox = new Box<>();
        Box<String> stringBox = new Box<>();
        Box<Double> doubleBox = new Box<>();

        System.out.print("Enter an Integer: ");
        integerBox.setValue(sc.nextInt());

        sc.nextLine();

        System.out.print("Enter a String: ");
        stringBox.setValue(sc.nextLine());

        System.out.print("Enter a Double: ");
        doubleBox.setValue(sc.nextDouble());

        System.out.println("\nStored Values");
        System.out.println("Integer : " + integerBox.getValue());
        System.out.println("String : " + stringBox.getValue());
        System.out.println("Double : " + doubleBox.getValue());
    }
}
```

```

        sc.close();
    }
}

```

5. TEST CASES

#	Input	Expected Output	Actual Output
1 (Typical Case)	100Java99.99	Integer : 100String : JavaDouble : 99.99	Same as Expected
2 (Edge Case)	0Hello World0.0	Integer : 0String : Hello WorldDouble : 0.0	Same as Expected
3 (Boundary Case)	-50Generics-45.75	Integer : -50String : GenericsDouble : -45.75	Same as Expected

6. SCREENSHOTS OF EXECUTION

Take **three screenshots**.

Test Case 1

Input

100
Java
99.99

Output

Enter an Integer: 100
Enter a String: Java
Enter a Double: 99.99

Stored Values
Integer : 100
String : Java
Double : 99.99

```

Enter an Integer: 100
Enter a String: java
Enter a Double: 99.99

Stored values
Integer : 100
String : java
Double : 99.99

```

Test Case 2

Input

0
Hello World
0.0

Output

Enter an Integer: 0
Enter a String: Hello World
Enter a Double: 0.0

Stored Values
Integer : 0
String : Hello World
Double : 0.0

```
Enter an Integer: 0
Enter a String: Hello World
Enter a Double: 0.0

Stored Values
Integer : 0
String : Hello World
Double : 0.0
```

Test Case 3

Input

-50
Generics
-45.75

Output

Enter an Integer: -50
Enter a String: Generics
Enter a Double: -45.75

Stored Values
Integer : -50
String : Generics
Double : -45.75

```
Enter an Integer: -50
Enter a String: Generics
Enter a Double: -45.75

Stored Values
Integer : -50
String : Generics
Double : -45.75
```

7. REFLECTION

Through this assignment, I understood how Java Generics help create reusable classes for different data types. I also learned how to use Scanner to accept user input and store different types of values using a single generic class.

8. DECLARATION

I hereby declare that this assignment is my own work. Any external resources or AI tools used were only for learning and understanding the concepts. The program was implemented, tested and verified by me before submission.